$$\begin{bmatrix} 1 & k \end{bmatrix}$$

$|A| = k^2 - 1$

$k^2 - 1 = 0$

$k = \pm 1$

Ans. • common value $\boxed{k = -1}$

WEEK-6

1. Linear Transformations: $T(U+V) = T(U) + T(V)$ (additivity)
$T(cU) = c\,T(U)$ (homogeneity)

2. Kernel & image

⇒ Kernel (Null Space of T): $\ker(T) = \{v \in V \mid T(v) = 0\}$
- always a subspace of V
- Nullity $(T) = \dim(\ker(T))$

⇒ Image (Range of T): $\mathrm{Im}(T) = \{T(v) \mid v \in V\}$
- always a subspace of W
- $\mathrm{Rank}(T) = \dim(\mathrm{Im}(T))$

• Rank-Nullity for LT: $\mathrm{rank}(T) + \mathrm{nullity}(T) = \dim(V)$

Q(1) Suppose $T: R^2 \to R^2$ is a linear transformation, such that $T(x,y) = (x,0)$. Which of the following options are correct?

Sol. ✓⇒ The matrix corresponding to T w.r.t. the standard ordered basis of R^2, i.e., $\{(1,0), (0,1)\}$, for both the domain and co-domain is $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

• use the standard basis: $e_1 = (1,0)$, $e_2 = (0,1)$

~~Computing, for~~ $\boxed{T(x,y) = (x,0)}$

$T(1,0) = (1,0)$

$T(0,1) = (0,0)$

T is neither one-one nor onto

$T(x,y) = (x,0)$

- if $x=0$, $\to (0,0)$ $\to$ whatever change in y Transformation is always 0, Not one-one X many-one

- $(x,0) \to$ just x-axis, not all of R^2

Not onto X

Q.2

- subspace:

$W = \{(x,y,z) \mid x = 2y + z\} \in R^3$

- Basis of W:

$\beta = \{(2,1,0), (1,0,1)\}$

- Linear transformation:

$T: W \to R^2$

with:

$T(2,1,0) = (1,0)$, $T(1,0,1) = (0,1)$

key idea: $(x,y,z) = a(2,1,0) + b(1,0,1)$

Step:1 Express (x,y,z) in terms of basis

$a(2,1,0) + b(1,0,1) = (2a+b, a, b)$

$x = 2a+b$, $y = a$, $z = b$

So: $a = y$, $b = z$

Step:2 Apply linearity of T

$T(x,y,z) = T[a(2,1,0) + b(1,0,1)]$

$= a \cdot T(2,1,0) + b \cdot T(1,0,1)$

Substitute: $\Rightarrow y(1,0) + z(0,1) \Rightarrow (y,z)$

$\Rightarrow$ appropriate definition of T: $T(x,y,z) = (y,z)$

- Domain W has dimension 2
- Co-domain W has dimension 2 } ~~one-one~~ onto
- Basis vectors map to linearly independent vectors ↳ one-one

so, T is Isomorphism

⇒ What will be the matrix representation of T wrt to the basis β for W and γ = {(1,1), (1,−1)} for R^2?

Sol. We want matrix of T w.r.t. :
- Basis β = {(2,1,0), (1,0,1)}
- Basis γ = {(1,1), (1,−1)}

T(2,1,0) = (1,0)
T(1,0,1) = (0,1)

Convert (1,0) into γ-coordinates

Solve: $(1,0) = a(1,1) + b(1,-1)$

$= (a+b, a-b)$ ⇒ $\boxed{a = \frac{1}{2},\ b = \frac{1}{2}}$

1 0

Solve: $(0,1) = a(1,1) + b(1,-1)$

$a + b = 0 \qquad a - b = 1$

$\boxed{a = \frac{1}{2},\ b = -\frac{1}{2}}$

Ans: $\frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

Q. Let β = {(1,0), (0,1)} and γ = {(1,1), (1,−1)} be 2 bases of R^2. If $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is the matrix representation of a linear transformation $T : R^2 \to R^2$ w.r.t β for both the domain and the co-domain. What is the matrix rep. of T w.r.t γ for the domain and β for the co-domain.

Sol. $\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} a+b & a-b \\ c+d & c-d \end{bmatrix}$

Kernel & Image Detailed

Q① Find the kernel of the linear transformation $T: R^3 \to R^2$ defined by $T(\vec{x}) = A\vec{x}$ with $A = \begin{bmatrix} 1 & -1 & -2 \\ -1 & 2 & 3 \end{bmatrix}$

Sol. $\boxed{T(\vec{x}) = A\vec{x} = \vec{0}}$ Imp. ★

$$\begin{bmatrix} 1 & -1 & -2 \\ -1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow \begin{matrix} x_1 = x_3 \\ x_2 = -x_3 \\ x_3 = x_3 \end{matrix}$$

$$\vec{x} = \begin{bmatrix} t \\ -t \\ t \end{bmatrix} = t \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$\ker(T) = \text{span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$$

Q.② Which of the following matrices corresponds to the given linear transformation T wrt the standard ordered basis for R3 and the standard ordered basis for R2.

Sol. ① Standard basis $R^3 \rightarrow \{(1,0,0), (0,1,0), (0,0,1)\}$

- $T(1,0,0) = [2(1)+3(0),\ 4(0)+0) \Rightarrow (2,0)$
- $T(0,1,0) = [2(0)+3(0),\ 4(1)+0] = (0,4)$
- $T(0,0,1) = [3,1]$

$\hookrightarrow [T] = [T(e_1) \cdot T(e_2)\ \ T(e_3)]$

$$= \begin{bmatrix} 2 & 0 & 3 \\ 0 & 4 & 1 \end{bmatrix}$$

---

② What is the basis of kernel of T?

Sol. $T(x,y,z) = (0,0)$

$(2x+3z,\ 4y+z) = (0,0)$

$2x+3z = 0 \qquad z = -4y$

$4y+z = 0 \qquad x = 6y$

$(x,y,z) = (6y, y, -4y) = -4y\left(\frac{6}{-4}, \frac{1}{-4}, \frac{-4}{-4}\right)$

$\Rightarrow \left\{\left(\frac{-3}{2}, \frac{-1}{4}, 1\right)\right\}$

(3) What will be the dimension of the subspace Im(T)

$$Im(T) = \begin{bmatrix} 2 & 0 & 3 \\ 0 & 4 & 1 \end{bmatrix}$$

Dim = No. of linearly independent rows/cols = ~~indep~~ rank

Ans. (2)

M-1 $col(1) \times \frac{3}{2} + col(2) \times \frac{1}{4} = col(3)$

so independent = 2

M-2 $A_1 V_1 + A_2 V_2 = (0,0)$

$A_1 (2,0,3) + A_2 (0,4,1) = (0,0,0)$

$(2A_1, 4A_2, 3A_1 + A_2) = (0,0,0)$

$\boxed{A_1 = 0}$ $\boxed{A_2 = 0}$ $A_3 \rightarrow$ no ean.

$\hookrightarrow$ Independent

---

(4) Ker ⇒ Spanned by vector (6, 4, -4)

$\boxed{ker \neq 0}$ ⇒ one-one ✗

Rank of T = Dimension of Im(T) ⇒ 2 R^2

Codomain = Range

R2 = R2

~~OTT~~ onto ✓

Q.3 Which option represents the kernel & image of the following linear transformation?

$T: R^2 \rightarrow R^2$

$T(x,y) = (x,0)$

Image ⇒ $T(\underbrace{x,y}) = (x,0)$ range of T

Sol. $T(x,y) = (x,0)$

$ker(T) = (0,0)$

$ker(x,y) = (0,0)$

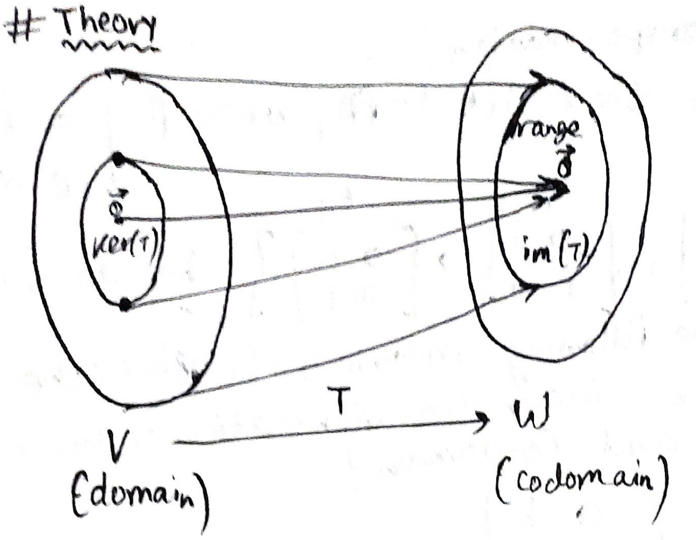
$(x,0) = (0,0)$

$x = 0$ (y) can be any thing because it doesn't appear in output

$ker(T) = Span\{(0,1)\}$

$Im(T) = Span\{(1,0)\}$

Theory



let $T: V \to W$ be a linear transformation.

- The kernel of T, ker(T), is the set of all input vectors $\vec{v} \in V$ such that $T(\vec{v}) = 0$
- Given $T(\vec{v}) = A\vec{v}$, the kernel of T is equivalent to the null space of A.
  - $ker(T) = \{\vec{v} \in V \mid T(\vec{v}) = \vec{0}\}$

- A linear transformation f is 1-1 if and only if $Ker(f) = 0$
- transformation $f: V \to W$ is onto if and only if $Im(f) = W$

- The image of T, im(T), is the set of all output vectors $T(\vec{v})$ such that $\vec{v} \in V$. im(T) = range of T
  Given $T(\vec{v}) = A\vec{v}$, the image of T is equivalent the column space of A.
  - $im(T) := \{T(\vec{v}) \mid \vec{v} \in V\}$

Points to Remember

- If v_1 and v_2 are in the kernel of a linear transformation $T: V \to W$, where V and W 2 vectors spaces, then $v_1 + v_2$ will also be in the kernel of T.
- Kernel of a linear transformation $T: V \to W$ is a vector subspace of V, where V and W are 2 vector spaces.
- If w_1 and w_2 are in the image of a linear transformation $T: V \to W$, where V and W are 2 vector spaces, then $w_1 + w_2$ will also be in the image of T.
- Image of a linear transformation $T: V \to W$ is a vector subspace of W, where V and W are 2 vector spaces.

Q. Consider the linear transformation: $M_{2\times2}(\mathbb{R}) \rightarrow M_{2\times2}(\mathbb{R})$ such that $T(A) = PA$, where $P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and a, b, c, d are in $\mathbb{R}$.

Let $B = \left\{ \underset{E_1}{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}}, \underset{E_2}{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}, \underset{E_3}{\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}}, \underset{E_4}{\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}} \right\}$ be an ordered basis of $M_{2\times2}(\mathbb{R})$. Which of the following matrices represents the matrix corresponding to the linear transformation T w.r.t β for both domain and co-domain?

Sol $T(A) = PA$ $\qquad P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

- $T(E_1) = \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a & 0 \\ c & 0 \end{bmatrix}$

Trick: $T(E_1) = aE_1 + 0E_2 + cE_3 + 0E_4$

- $T(E_2) = \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & a \\ 0 & c \end{bmatrix}$

$T(E_2) = 0E_1 + aE_2 + 0E_3 + cE_4$

$T(E_3) = bE_1 + 0E_2 + dE_3 + 0E_4$

$T(E_4) = 0E_1 + bE_2 + 0E_3 + dE_4$

$$\begin{bmatrix} a & 0 & b & 0 \\ 0 & a & 0 & b \\ c & 0 & d & 0 \\ 0 & c & 0 & d \end{bmatrix}$$